

3 Partial Derivatives

3.3 Partial Derivatives

Definition 3.3.1 If $f(x, y)$ is a function of two variables, then the partial derivative of f with respect to x at a point (a, b) is given by

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}.$$

Similarly, the partial derivative of f with respect to y is given by

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

When these derivatives exist.

Example 3.3.2 Find the first partial derivatives of the function

$$f(x, y) = x^3 + x^2y^3 - 2y^2.$$

Find $f_x(2, 1)$.

Example 3.3.3 Find the first partial derivatives of the function

$$f(x, y) = \sin\left(\frac{x}{1+y}\right).$$

Interpretation of partial derivatives:

Example 3.3.4 Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if

$$x^2 + y^2 z + 6xyz - z^2 + 10 = 0.$$

Example 3.3.5 Find the partial derivatives of the function

$$f(x, y) = \sin(x) \cos(y)$$

Example 3.3.6 Find the domain and the partial derivatives of the function

$$f(x, y) = \sqrt{x - y^2}$$

3.3.1 Higher Derivatives

Example 3.3.7 Find the second partial derivatives of

$$f(x, y) = x^3 + x^2y^3 - 2y^2.$$

Example 3.3.8 Find $f_{xxyz}(x, y, z)$ if

$$f(x, y, z) = \sin(3x + yz).$$

Theorem 3.3.9 *Suppose that f is defined on a disk D that contains the point (a, b) . If the functions f_{xy}*

and f_{yx} are both continuous on D , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$